

Data Analytics

Research Assignment 1.

1. What is a Database? Define it in your own words and explain why databases are essential in the modern world. Provide two real-world examples of where databases are used.

A Database - is a place where data is stored in an organized way where you can easily access it.

Why databases are essential in the modern world.

- > They handle large amount of data.
- > fast to search
- > Many people can use it at once
- > Database keep data safe.

Examples.

Banking system

- Database is used to store account numbers, transaction history and balances.

Hospital system

- Database is used to store patients records, appointments and medicine.

Source - Chat Gpt

IBM (2023)

Q2 what is a Data warehouse? How does it differ from a regular database? why would a company use a data warehouse.

A data warehouse - is a centralized system designed to store large amount of structural data from multiple sources.

Difference between Database & Data warehouse

<u>Database</u>	<u>Datawarehouse</u>
- Designed for (OLTP) Online Transaction processing. - Daily running of the business - current real time data	- Built for (OLAP) Online Analytical processing. - for reporting, combines data from multiple databases. - Analyze historical data

why would a company use a Data warehouse

- To consolidate scattered data
- Enable fast, complex analysis
- Protect operational systems
- Support historical trending
- Improves data quality and consistency

Source

inman W.H (2005) Building the Data warehouse (4th ed). Wiley
Oracle Corporation. (2023) Database Data warehouse guide (26c) Oracle help center.

Q 3. What is a data lake? Explain the concept and describe how it differs from a data warehouse, when would a business prefer a data lake over a data warehouse?

A Data lake is a central storage system that allow a company to store large amounts of data in its original form. The data can be structured data like tables, spreadsheets, and unstructured data like images, videos and audio.

<u>Data lake</u>	<u>Data warehouse</u>
Structured, semi -unstructured Raw data Stores all website logs for future analysis	Structured data cleaned data, stores monthly sales figures for management report.

A business would prefer a Data lake when it need to store very large amounts of diverse data.
Example.

- When the business have many different type of data.
- Uncertain about the future of data.
- Extremely large volume of data

Source

IBM (2025)

Q.4. What is data mart? How does it relate to a data warehouse and who typically uses it.

A data mart is a smaller subset of a specific database that contains data focused on one department extracted from the main data warehouse.

> Data mart is built from data warehouse.
Data mart depend on data warehouse.

> Data comes from different sources and goes into data warehouse. From data warehouse you only select specific data and create data mart.

Who typically uses it.

> Specific departments or business units within an organization.

> Marketing team

source IBM (2003)

Q5. Explain the difference between structured, semi-structured and unstructured data. provide one example of each.

Structured data

- Data is organized, easy to search.
- Fits into tables with rows and columns.

Example

Customer-table

customerID	customer name	Age
101	Boltunpio	30
102	Luftuno	35

Semi structured data

- Data that is not in a perfect table
- Does not fit neatly into rows and columns

Example,

Emails

Unstructure data

> Data that is hard to organize in rows / column

Example

WhatsApp voice notes, videos, photos, emails.

Source

Oyode (2024)

What is Data Modeling? Why is it important when designing a system that works with data?
Describe at least two types of data models.

Data modeling - is a process of planning and organizing data in a system.

Why is it important when designing a system that works with data?

It's important because it organizes data, Reduces duplication, improves data retrieval and support business requirements.

Two types of data models.

- > Relational data model - organizes data into table ^(consistency of rows & columns)
- > Hierarchical data model - Organizes data in a tree-like structure, data has parent and child relationships.

Example of relational data models

Student-ID	Student-name	Course-ID	Course name
2102565	Thato	BTFS	Information system
2102645	Nawato	NDPAC	Accountancy

Example of hierarchical data model

University

→ faculty → Department → Students

Source

Cotsonel, C & morris S. (2019)
Database systems: Design Implementation,
t Management. 13th ed. Cengage Learning.

What is Data Mining? How is it used in business?
Provide two real-world examples of data mining in action.

Data mining - analyzes large amounts of data to discover hidden patterns and relationships that help businesses make better decisions

How is it used in business?

- > To Predict the future - what will customers buy
- > To understand customers - who are my best customers
- > To stop losses. - Detecting customers who will leave

Examples.

Shopify - mines your buying history

Bank - mine millions of transactions.

Vodacom - Check which customers are about to leave and give them a special to stay.

Q. What is DBMS? Name and briefly describe four popular DBMS platforms and explain a suitable use case for each.

DBMS - is a software that creates, stores and manages data in an organized way.

Four popular DBMS platforms

① MySQL - Most popular open source, relational database use case - Facebook, small online shops.

② PostgreSQL - Free and open-source. Can handle complex data and big data.
use case - Banking systems, data analysis

③ Oracle Database - Very powerful, commercial DB for huge enterprises.
Best for banks with millions of transactions.
use case - FNB SARS - when data is huge and you need top security and support.

④ MongoDB - flexible - store data as JSON documents, not tables.
Best for modern apps where data changes a lot.
is not fixed.
use case - social media apps.

Q1. What is ETL (Extract, Transform, Load)
explain each step and describe why this process
is important in data analysis.

Extract, Transform and Load - is a process that extracts data from different sources, cleans and transforms it into a standard format and then loads it into a central data warehouse for analysts.

Three Steps

1. Extract - collect raw data

2. Transform - Data is cleaned.

3. Load. - Clean data is transferred to the data warehouse.

ETL is important in Data Analytics because of the following four points.

- > One version of Truth - combines messy sources into 1 clean trusted place.
- > Better Decisions - clean data = clean reports
- > Dirty data = wrong decisions.
- > Saves Time - Analysts don't waste time cleaning they can just analyze.
- > Supports Data Mining - You can't mine data on dirty data.

Question 10

What are common data storage formats used in analytics?

Briefly describe each and state when you would use one over another.

→ Data storage format - is how data is physically saved on a disk.

Format used in analytics.

1. CSV (Comma Separated values)

- Simple text file separated by commas.

- Everyone can open it

- It is very slow

→ You can use ^{CSV} when you are sharing small data.

2. JSON (JavaScript Object Notation)

- JSON - Data with keys and values

- Use when working with APIs, web apps, MongoDB or data with changing structure.

3. Parquet

→ Stores data by columns not rows.

- Highly compressed

- Very fast for analytics

- Can't be read by humans

→ Use Parquet format when you have big data and want to run SQL fast.

4. Avro

→ Stores schema with data

→ It is great for writing data quickly

- Slower for reading analytics

use when data is being moved.

S. ORC (Optimized Row Columnar)

- It is column based

- Super fast, great compression.

use when you are using hadoop

- when you need to read and analyze fast

Source Databricks (2024) Parquet vs Avro

Question 11

What is Artificial Intelligence? Define it and explain the difference between Narrow AI and General AI. provide one example of each.

Artificial intelligence is when computers are made to think, learn and make decisions like a human being.

Difference between Narrow AI and General AI

- Smart at a specific task
- Does it one job better than human.

example

Chat GPT

Capital fraud detection

General AI

- Can understand anything like a human being

example

A Robot that can talk

Source

IBM (2020)

Britannica. (2024)

Question 12.

What is machine learning? How does it differ from traditional programming? Describe the three main types of machine learning: supervised, unsupervised and reinforcement learning.

Machine learning (ML) - computers are given data and they learn the pattern by themselves to make predictions.

Traditional Programming
You tell the computer exactly what to do.

Machine learning - you don't write rules.

- you show it multiple examples and it will figure it out the rules.

- learn rules from data.

Types of Machine learning

1. Supervised learning

- you give computers data that is already labeled and it learns to predict.

example - predicting house prices

2. Unsupervised learning

- computers must find hidden patterns by itself.

example - customer segmentation

3. Reinforcement learning

- computer learns by doing something and getting a reward or punishment.

- Data is not given, it learns by playing
example - self driving car.

Question 13

What is deep learning? How does it relate to machine learning and Artificial Intelligence? What are neural networks and how do they work?

Deep learning - a powerful part of machine learning that uses brain-like networks to learn from very large amounts of data.

Deep learning is a subset of machine learning, which is a subset of Artificial Intelligence.

Neural Network is a network of small decision makers called neurons connected like brain cells.

How neural networks work.

1. Input layer - The eye

- Receives raw data

2. Hidden layers - The Brain

- Each layer passes what it learned to the next layer

3. Output layer - The mouth

- Gives final answer.

∴ Input > Hidden > output

Britannica (2024) Source.

Question 14

What is Natural Language Processing (NLP)? Describe what it is and provide three real-world applications of NLP that you use in everyday life.

Natural Language Processing - is a technology that allows computer to understand human language.

NLP is the translator between human language and computer language.

Real world applications

1. WhatsApp

2. Chat GPT

3. Social media (TikTok)

Source: AWS (2024)

Question 15

What is a large language model (LLM)? How is it different from a standard machine learning model? name two well-known LLMs and explain what they are used for.

Large Language Model (LLM) - Is an advanced AI that has read almost the entire internet so it can understand and write like a human.

How is LLM different from a standard machine learning

Standard Machine learning is narrow and task specific while large language model is general purpose and understands language

Two well known LLMs and what used for

1. Chat GPT

- used for writing essays, summarizing notes, answering questions, creating images.

2. Gemini (Google)

- used inside Google search, Gmail to summarize emails, Google documents to help you write

Source

IBM (2024)

IBM.com

Question 16

What is generative AI? How does it work at a high level? Name three tools that use generative AI and describe what each one does.

- Generative AI is Artificial Intelligence that can create new things like text, images, music, and videos that never existed before, like a human creator.

How does it work at a high level?

- > Training on huge data
- > It learns the pattern
- > Generating new content.

Three tools that use generative AI.

1. ChatGPT - generates text.

2. Canva AI

- generates new images from text.

3. Suno AI

- generates new song from text.

Source: Google Cloud (2024)

Question 17

What is a Training Dataset? Why is the quality of training data so important when building AI model? What could go wrong if the training data is biased or incorrect.

Training data - Is the examples shown to a computer so it can learn.

Why is the quality of training data so important?

- > Accuracy depends on data.
- > Model can only learn what you show
- > Costly to fix later.

What could go wrong if the training data is biased or incorrect?

> Biased

- AI becomes racist

Example if you train facial recognition only with black faces, it will fail to recognise white faces in midland.

> Incorrect.

- AI makes dangerous mistakes.

Example if bank transaction data has wrong labels like fraud marked as not fraud. AI will allow fraud to steal money.

Source

developers.google.com.

Question 19

What is computer vision? How do machines "see" and interpret images? provide two real world use cases of computer vision.

Computer vision is a technology that allows computers to see and understand pictures and videos.

How do machine see and interpret images?

- > Images as numbers
- > feature extraction using deep learning
- > Interpretation

Use case

1. Facial Recognition - unlock your phone

- your phone uses computer vision to scan your face and unlock.

2. Security Cameras

- Shoprite uses CCTV that counts people and detects shoplifting.

Source.

IBM (2024)

AWS (2024)

aws.amazon.com

Question 14

What is an API (Application Programming Interface)?

Explain what it is using a simple real-world analogy.

Why are APIs important in modern software and data systems?

Application Programming Interface is a messenger that takes your message from one app to another and come back with an answer.

Simple real world analogy

> Restaurant waiter analogy

Customer = App (example banking app)

Kitchen = Another system

Menu = API Documentation

Waiter = API

"You tell" waiter you want burger, waiter goes to the kitchen, tell the chef, chef prepares burger, waiter bring back the burger to you.

Why API is important in modern software and data systems

- > Connects different systems
- > Saves Time and money
- > enable data sharing securely
- > Powers everything you use daily.

APIs are essential for connecting unstructured sources and AI models into one working pipeline without exposing the whole system.

Source IBM (2024)
AWS (2024).

Question 20

What is an API key? What is its purpose? How should it be stored and protected and what could happen if it is exposed publicly?

Application Programming Interface It's a unique code. It proves who you are and gives you permission to enter and use the API.

API Purposes

- Authentication - ~~controls what you can access~~
 - Proves you are a registered user
- Authorization
 - Controls what you can access

Tracking

- Track usage per key.

How to Store

Store in environment variables, not in code.

- Use Secret Manager
- Never Share on GitHub / Email
- Rotate regularly & Limit Permissions

What could happen if it is exposed.

if you post API key on GitHub

- Hackers will steal
- Financial Loss
- Data Theft

If exposed regenerate immediately the key in dashboard

Question 21

What is a REST API? What does REST stand for and what are the main HTTP methods used in REST APIs? Briefly describe each method.

REST stands for Representational State Transfer. REST API is a way of building APIs that uses the same methods your browser uses to browse the Internet to exchange data.

The main HTTP methods used in REST APIs are:

GET

POST

PUT

DELETE.

1. GET - Retrieve Data

- GET data from server

2. POST - Create New Data

- send new data to server to create something

3. PUT - Update existing data.

4. DELETE - Remove Data.

Source

AWS (2024)

aws.amazon.com

Question 22

What is JSON (Javascript Object Notation)? Why is it commonly used when working with APIs and data exchange? Write a simple example of a JSON object.

JSON is a way to write data that both humans and computers can easily read.

Why is JSON commonly used?

1. Easy to read for humans and computer.
uses less data
2. Language independent
3. Perfect for REST APIs
4. Works with Structured and Semi Structured Data.

Example of a JSON object.

Rule - "always use double quotes" not single'

Simple Student

```
[ "Student-id": 120,  
  "name": "Boltumero",  
  "Surname": "Seteto",  
  "Course": "FIS",  
  "is_active": true,  
  "year": 2016  
]
```

aws.amazon.com

Question 23

What is prompt Engineering? Why is it an important skill when working with AI tools like ChatGPT or Claude? Provide two examples of a poorly written prompt and an improved version of each.

Prompt Engineering is learning how to ask AI the right question in the right way, so it gives you the right answer.

Why is it an important skill when working with AI tools

1. Garbage in, Garbage out (GIGO)
- poor prompt = poor answer
2. Saves Time and money
3. Reduces Bias and Hallucination
4. It is a job skill

Examples of poorly written vs Improved

Poor Prompt

Tell me about AI

Improved prompt

Explain Artificial Intelligence in simple terms for a grade 12 student. Give definition, narrow and general and an example, use bullet points

Prompt engineering bridges the gap between human and narrow AI capabilities by turning unstructured human request into structured instructions that leverage the model training data effectively.

5. AWS (2024) what is prompt engineering

Question 24

What is a Map function? Explain what it does in programming and data processing. Provide a simple example in either Python or JavaScript.

➤ Map is a function that takes a list of items and applies the same operation to every item in the list and returns a new list with the changed items.

➤ It's a pure function that supports functional programming and is ideal for parallel processing of structured data before model training.

- shorter code, faster, clean

Example

make all student names uppercase.

Python code.

```
names = ("Botumelo", "Lerato", "Thendo")
```

```
names = list(map(lambda name: name.upper(), names))
```

```
print(upper_names)
```

```
# Output: ["BOTUMELO", "LERATO", "THENDO"]
```

Source, Mozilla. (2024)

Array.prototype.map

developer.mozilla.org

W3Schools. (2024)

W3Schools.com

Question 25

What is cloud computing? name the three main cloud service models and briefly explain each, Give one example. Give one example of a product for each model.

Cloud computing is using computer, software, storage over the internet ~~without~~ instead of buying your own.

Three main cloud services models

IaaS - you buy ingredients - (you manage everything)

PaaS - They manage half - (you buy pizza kit, they give you)

SaaS - They manage everything (you order pizza, they deliver)

IaaS - Infrastructure as a Service.

- you rent raw IT infrastructure.

example - IIC rents 10 servers from AWS EC2

PaaS - platform as a service

~~- you rent raw IT infrastructure~~

- you rent a platform where you can just deploy your code without worrying about servers.

example - Developer writes python flask code and deploys on Heroku

SaaS - Software as a Service

- you use software via browser, you don't build anything.

example - we use Gmail to send email.

Question 2b

What is a version control system? What is Git and why is it widely used by data analysts and developers? Briefly explain what a commit and a branch are.

Version Control System is a software that track and saves every change you make in your files. It saves every version of your project.

Git is the most popular version control system in the world

Why is it widely used by data analyst and developers

1. Teamwork
2. History & undo
3. Backup in cloud and traces who did what
4. Industry standard for jobs

Commit - Save

Commit is like when you press save with a message.

Branch - copy

Branch is a separate copy of your project where you can work on new feature without breaking main working code.

Source.

alt (2024)

IBM (2024)

Question 27

What is a Jupyter Notebook? What programming language is it most commonly used with? Why is it popular among Data Analysts and Data Scientists?

Jupyter Notebook is like MS Word and calculator and graph paper combined for coders. you write a little code, run it instantly, see graph, write explanation below - all in all same page.

Programming language is commonly used by python

Why is it popular among Data Analysts and Data Scientists?

1. Run code in small pieces - instant feedback
2. code + notes + graphs in one place
3. Perfect for Data Exploration & Visualization
4. Easy to share and present
5. Supports libraries for AI and Data

Source: Jupyter (2024)

QUESTION 24

What is python? Why is it considered one of the most popular programming languages for Data Analytics and AI? Name four python libraries commonly used in data work and briefly describe each.

Python is a high-level - easy to read and write. and it can do almost anything.

Why is it considered one of the most popular programming languages for Data Analytics and AI?

- Simple to write and read.
- Huge libraries for Data & AI
- Perfect for jupyter notebook
- Handles all data types
- Biggest community on Stack overflow.

Four python libraries commonly used in data work.

1. pandas

- Data Cleaning & Analysis like excel in python. Works with tables.

2. Numpy - maths & numbers - fast calculations on big lists

3. matplotlib/seaborn - graphs & visualization - make charts, line graphs, bar graphs.

4. Scikit-learn - AI & machine learning - train AI models, detect fraud, predict.

Source, Python.org (2024)

Question 29

What is Data privacy? Why is it important? Research the GDPR (General Data Protection Regulation) and explain what it is and which countries it applies to.

Data privacy is the right of a person to control their personal information - who can collect it and share it.

Why is it important?

- Prevents identity Theft & fraud.
- Stops Discrimination & Bias
- Legal Requirement
- Build Trust for AI & cloud

General Data protection Regulation (GDPR)

Is the Strictest data Privacy law in the world.

The GDPR requires that personal data be:

- Processed lawfully, fairly and transparently
- collected for legitimate purposes
- minimized
- accurate
- Stored no longer than needed.

Key rules of GDPR

- consent, Right to Access, Right to be forgotten.

created by the European Union

GDPR applies to

Austria, Belgium, Bulgaria, Spain, Sweden, Norway, Iceland, Denmark, Germany, Netherlands, Poland, Greece

GDPR EU (2024)

Question 3

What is Data ethics? provide three examples of unethical use of data or AI, and explain what makes each example problematic.

Data Ethics - is the rules and principles about what we should and shouldn't do with data and AI.

Examples of unethical use of data or AI

- Biased AI Recruitment Tool

what makes it problematic / unethical?

- Violates Fairness & non Discrimination
- Lack of Transparency
- Feedback loop

2. Facial Recognition Selling without Consent.

- > Violates privacy & GDPR
- > mass Surveillance
- > Inaccuracy for Black people

3. Using Health App Data to Increase Insurance prices.

- Violates purpose Limitation
- Exploitation of Vulnerable
- no Accountability

Source - Dastin, J (2018)

Question 31

What is Data Governance? Define it and explain the key components of a good data governance framework

Data Governance - is who can touch data, how they touch it and making sure data is clean, safe and legal - like a constitution for data.

key components of a Good data governance framework

> Data Quality Management

- ensure data is accurate complete and not duplicate

> Data Security & Privacy

- protect personal info from hackers.

> Data Roles & ownership

- clear who is responsible

> Data policies & standards. - written rules.

> Compliance & Auditing - Track every change via Git com

Source Microsoft (2024) Data Governance Framework
learn.microsoft.com.

Question 32

What is Bias in AI? Explain what it means for an AI model to be biased. provide real world example of AI bias and describe the potential consequences.

Bias means AI learned human prejudices from bad training data and now makes unfair decisions - like rejecting all women.

What it means for an AI model to be biased.

- Means model does Not treat all groups equally.
Its accuracy is high for one group and low for another it is not neutral.

Real world Example of AI Bias

→ AI for Criminal Justice in USA (2016)

US courts used AI called COMPAS to predict if Criminal will re-offend and decide bail

Bias found that black defendants were 2x more likely to be labeled High risk of re-offending even when they did not re-offend.

Potential Consequences:

- unfair imprisonment and loss of freedom
- loss of Trust in AI and law
- legal violation

Source: Angwin J. et al (2016)

Question 33

What are the key differences between a Data Analyst, a Data Scientist, a Data Engineer and an AI Engineer? Create a Simple table comparing their roles, skills and tool.

Data Analyst - Answers what happened?

Data Scientist - Answers why happened and what will happen?

Data Engineer - Builds Pipelines so data is available

AI Engineer - Builds AI models into real products

	Data Analyst	Data Scientist	Data Engineer	AI Engineer
Role	Analyze data to find trends / make reports / dashboards	Build predictive models, find patterns	Collect, clean, move data from API	Take AI model and deploy as AI
Skill	Excel, SQL	python	python, SQL	python,
	Data cleaning Storytelling	Stats, machine Learning, Jupyter	ETL, cloud, git	Deep learn Cloud pro
Tool	Excel, Power BI, Tableau	Jupyter, pandas numpy, matplotlib	SQL, AWS Glt, GCP,	python Fast API
	SQL	lib, scikit-learn	Airflow	

Question 34:

What is Business Intelligence? Name two popular BI tools and explain how they help businesses make data-driven decisions.

BI is the process, tools and technologies that turn raw data into useful information, dashboards and reports so managers can make data driven decisions.

Two Popular BI Tools

1. Power BI

- > Drag and Drop Dashboards
- > Real-time Alerts
- > Cheap and integrates with Excel

2. Tableau

How it helps businesses

- Visual Exploration
- Self service
- AI inside.

Question 35

Which concept from this assignment did you find most interesting or surprising? Explain why?

Question 32 was most interesting in Bias in AI. I thought AI was neutral but it's not. AI doesn't fix human unfairness. It scales it to millions via cloud.

Question 3b.

Which concept did you find most difficult to understand? What resources did you use to help you understand it better?

The concept I found most difficult to understand initially was REST API - Specifically the difference between API, REST API, API key and JSON

> Too many new terms that sound the same.

Resources used

Youtube Visual Explanation

- What is API in 5 minutes

Question 37

How do you see Data and AI affecting your future career or everyday life? Be specific with examples.

I see Data and AI affecting me in two ways

- Everyday life and future career
- Everyday life
 - money & fraud protection
 - I use capitec app daily. AI ~~for~~ fraud detection already checks my transactions in real time.

Future career

I want to be a Data Analyst

As a Data Analyst in next 10 months

companies in midrand don't need more excel they need BI

my job will be: pull data from company REST

APIs secured with API keys

- clean it with python map

- push clean version with git commit / Branch

- Build dashboard in Power BI

- Ensure all data follow Governance framework.

without python, SQL, Power BI and cloud skills it will be unemployable. Every job asks for this skills.

References.

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